

Final project.

Introduction

Bacterial genome annotation links raw DNA sequence to biological interpretation by identifying genes, predicting their products, and reconstructing genome organization such as operons and regulatory elements. Reliable annotation typically requires combining automated pipelines with manual curation, because ab initio gene prediction, functional inference from homology, and regulatory motif detection can be uncertain—especially for poorly characterized lineages.

In this project, a 20,000 nt genomic fragment was analysed from an uncharacterized bacterium belonging to the newly proposed phylum Candidatus Kapaibacteriota (FCB group; Bacteroidota/Chlorobiota group; Kapaibacteriota sp. OB-3-ph). The main goal was to produce a publication-style annotation of this region by identifying coding and non-coding genes, assigning putative functions (with particular attention to hypothetical proteins), and describing gene organization.

Beyond basic feature calling, the analysis was designed to place the region into a broader biological context by defining operons as co-linear genes separated by short intergenic regions, examining long operons for possible regulatory sequences, screening for candidate horizontal gene transfer events using taxonomic signals from sequence similarity and phylogenetic analysis, and evaluating whether the region contains genes suggestive of secondary metabolite biosynthesis. Together, these steps aim to characterize the functional potential and evolutionary history of the studied genomic locus in a bacterium with limited prior annotation resources.

Results

Annotation using Prodigal, Glimmer3, GeneMarks-2.

Prodigal predicted 23 CDSs across the 20 kb fragment. Of these, 21 genes are supported by at least one of the alternative predictors (Glimmer3 or GeneMarkS-2), whereas two short Prodigal ORFs (genes 1_12 and 1_22) were not recovered by either tool and were therefore considered low-confidence candidates.

Most genes were strongly supported: 18 of 23 Prodigal CDSs were predicted by all three tools with nearly identical coordinates (e.g. 1_3–1_10, 1_13–1_14, 1_17–1_21), and can be considered high-confidence coding sequences. A small subset of loci showed either missing support in one caller or noticeable boundary discrepancies; these are the main candidates for manual curation in subsequent steps.

Automated annotation with Prokka (functional overview).

Using PROKKA, 18 protein-coding sequences (CDS) were identified within the fragment, with no tRNA or rRNA genes detected. Among these 18 CDSs, 10 were assigned specific functional product names, whereas 8 were annotated as hypothetical proteins, indicating that roughly half of the predicted proteins currently lack a clear functional characterization in available reference databases.

These results were used as an additional source of functional hypotheses and standardized product naming, while disagreements between Prokka and the multi-tool consensus gene set were treated as targets for manual inspection.

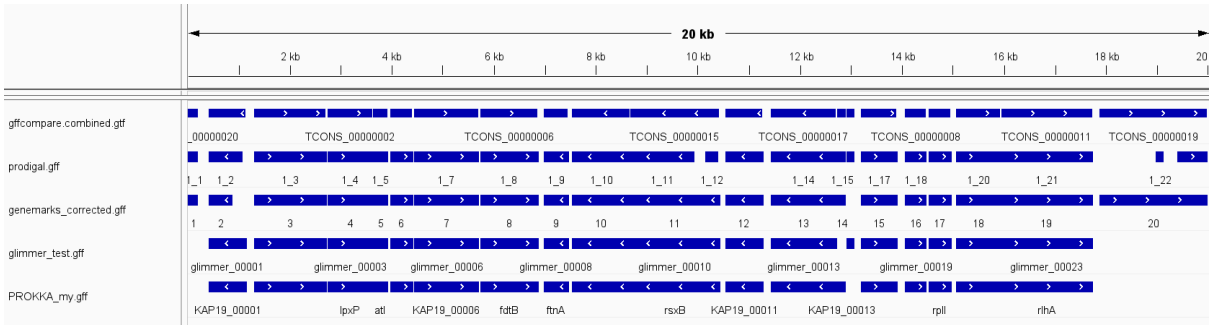


Fig. 1 IGV alignment of Prodigal, Glimmer3, GeneMarkS-2, their gffcompare combined and PROKKA annotations.

Operon organization.

The first operon longer than two genes (KAP19_00002–KAP19_00007) consists of six co-directional CDSs with intergenic distances from –8 to 14 bp, including LpxP and FdtB-like enzymes, and likely represents a cell-envelope and sugar-modification locus.

Applying an operon definition of co-directional genes with intergenic distances < 100 bp, three operons were identified: a six-gene plus-strand operon (KAP19_00002–KAP19_00007) involved in cell-envelope and sugar modification, a five-gene plus-strand operon (KAP19_00014–KAP19_00018) enriched in translation/RNA-modification functions, a four-gene minus-strand operon (KAP19_00008–KAP19_00011) related to redox and metal homeostasis, and a two-gene operon (KAP19_00012–KAP19_00013) with an unknown function, while the remaining CDSs appear as single-gene transcriptional units.

A more detailed description of identified operons is included in the Methods section.

Regulatory sequence analysis for long operons.

Regulatory regions were examined for operons longer than four genes, focusing on the upstream region of the long operon (KAP19_00002–KAP19_00007). Approximately 250 bp upstream of the first gene of the long operon were inspected manually for σ^{70} -like promoter motifs. Several T-rich hexamers resembling –10 boxes were observed, but no clear, canonical TTGACA/–35 and TATAAT/–10 pair with appropriate spacing could be unambiguously identified, and no recognizable palindromic operator-like sequence was evident. In view of the phylogenetic distance of Candidatus Kapaibacteriota and the absence of an experimentally supported promoter model, no specific upstream regulatory region can be confidently assigned for this operon.

To evaluate transcription termination signals, ~200 bp downstream of the last gene of the KAP19_00002–KAP19_00007 operon were inspected for GC-rich inverted repeats followed by T-tracts characteristic of Rho-independent terminators. In addition, a TransTermHP-based web service was used to search for terminator sequences. Although a few short palindromic

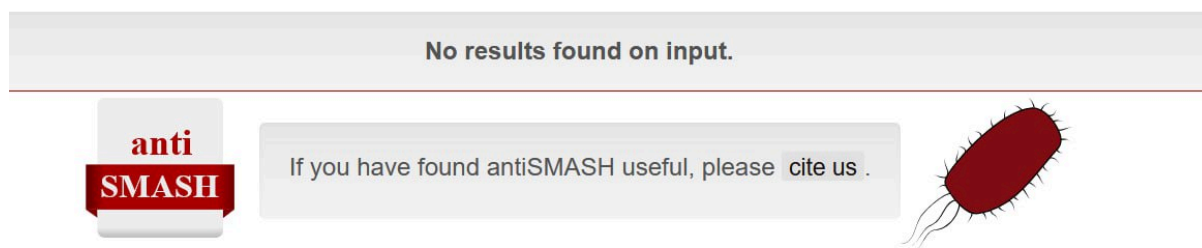


Fig.3 antiSMASH output.

Methods

ORF prediction using Prodigal v2.6.3, GeneMarkS-2, Glimmer3. Gffcompare.

Coding sequences were predicted using three ab initio prokaryotic gene finders: Prodigal, Glimmer3, and GeneMarkS-2. Prodigal v2.6.3 was run on the provided genomic fragment (`fragment_19.fasta`) with default single-genome settings, producing a GFF3 annotation file (`prodigal.gff`).

Open reading frames were then predicted de novo with Glimmer3 using the `g3-from-scratch.csh` pipeline on the same input sequence. The resulting ORF predictions were converted to GFF3 format with a custom `awk` script that: (i) recoded the contig name to `NODE_1_length_974525_cov_233`, (ii) ensured all features were typed as `CDS`, (iii) normalized coordinates and strands, (iv) set the phase field to 0, and (v) assigned stable IDs of the form `glimmer_XXXXX`. A GFF3 header line (`##gff-version 3`) was then prepended to yield the final Glimmer annotation file (`glimmer_test.gff`).

GeneMarkS-2 predictions were obtained in GFF3 format using the bacterial model, but the original output contained paired `gene` and `CDS` features with identical coordinates for each locus. To make this output compatible with downstream comparison tools, `genemarks.gff` was manually filtered so that only `CDS` features were retained and all `gene` feature lines were removed, producing `genemarks_corrected.gff`.

Comparison of annotations by different tools and consensus annotation.

To compare the predictions from different tools, gffcompare v0.12.10 was run with the Prodigal GFF3 file as the reference annotation and the Glimmer and GeneMarkS-2 GFF3 files as query sets. This generated summary statistics for each tool relative to Prodigal and a combined consensus GTF file used in subsequent steps of the analysis.

```

PATH=$PATH:/home/Shared/tools/Prodigal
prodigal -i fragment_19.fasta -d prodigal.fna -f gff -o prodigal.gff

PATH=$PATH:/home/Shared/tools/glimmer3.02/bin:/home/Shared/tools/glimmer
3.02/scripts
g3-from-scratch.csh fragment_19.fasta glimmer3
grep ^orf ./01_gene_calls/glimmer3.predict | awk '{OFS="\t"}{strand =
"+"; start = $2; end = $3} {if($4 < 0) {strand="-"; start = $3; end =
$2}}{gsub(/[\+-]/," Bioinformatics course, Skoltech, 2025 ")}{print
"region", "GLIMMER", "CDS", start, end, $5, strand, $4, "ID="$1";
NOTE:GLIMMER ORF prediction;"}' > glimmer3_to_gff.gff

# Convert glimmer
awk 'BEGIN{OFS="\t"} {
    $1 = "NODE_1_length_974525_cov_233";
    $3 = "CDS";
    split($9, parts, ";");
    split(parts[1], id, "=");
    gsub("orf", "", id[2]);
    $9 = "ID=glimmer_" id[2];
    print $1, $2, $3, $4, $5, $6, $7, "0", $9;
}' glimmer3_to_gff.gff > glimmer_improved.gff

# Add title
echo "##gff-version 3" > glimmer_test.gff
cat glimmer_improved.gff >> glimmer_test.gff

##### gffcompare #####
gffcompare -r prodigal.gff -o gffcompare glimmer_test.gff
genemarks_corrected.gff

```

Prodigal predicted 23 CDSs across the 20 kb fragment. Of these, 21 genes are supported by at least one of the alternative predictors (Glimmer3 or GeneMarkS-2), whereas two short Prodigal ORFs (genes 1_12 and 1_22) were not recovered by either tool and were therefore considered low-confidence candidates.

Table 1. Consensus-support between 3 annotation tools: Prodigal, Glimmer3 and GeneMarks-2.

Prodi gal ID	Coordinates (strand)	Glimmer ID	GeneMar k ID	In Prodiga l	In Glimme r	In GeneMar k
1_1	1–240 (-)	–	1	yes	no	yes

1_2	404–1072 (–)	glimmer_0000 1	2 (404–889)	yes	yes	yes (shorter)
1_3	1297–2736 (+)	glimmer_0000 2	3	yes	yes	yes
1_4	2733–3641 (+)	glimmer_0000 3	4	yes	yes	yes
1_5	3634–3951 (+)	glimmer_0000 4	5	yes	yes	yes
1_6	3961–4419 (+)	glimmer_0000 5	6	yes	yes	yes
1_7	4421–5713 (+)	glimmer_0000 6	7	yes	yes	yes
1_8	5728–6894 (+)	glimmer_0000 7	8	yes	yes	yes
1_9	6961–7473 (–)	glimmer_0000 8	9	yes	yes	yes
1_10	7531–8682 (–)	glimmer_0000 9	10	yes	yes	yes
1_11	8658–9938 (–)	glimmer_0001 0 (8658–10445)	11 (8658–10 436)	yes	yes (longer)	yes (slightly shorter)
1_12	10136–10408 (+)	–	–	yes	no	no
1_13	10526–11302 (–)	glimmer_0001 1	12	yes	yes	yes
1_14	11427–12728 (–)	glimmer_0001 3	13	yes	yes	yes
1_15	12725–12922 (–)	–	14	yes	no	yes
1_16	12923–13102 (–)	glimmer_0001 7	–	yes	yes	no
1_17	13175–13918 (+)	glimmer_0001 8	15	yes	yes	yes

1_18	14039–14491 (+)	glimmer_0001 9	16	yes	yes	yes
1_19	14524–14976 (+)	glimmer_0002 0	17	yes	yes	yes
1_20	15038–15943 (+)	glimmer_0002 1	18	yes	yes	yes
1_21	15936–17750 (+)	glimmer_0002 3	19	yes	yes	yes
1_22	18981–19151 (–)	–	–	yes	no	no
1_23	19384–19998 (+)*	–	20 (17863–19998)	yes	no	yes (extends upstream)

Most genes were strongly supported: 18 of 23 Prodigal CDSs were predicted by all three tools with nearly identical coordinates (e.g. 1_3–1_10, 1_13–1_14, 1_17–1_21), and can be considered high-confidence coding sequences. A small subset of loci showed either missing support in one caller or noticeable boundary discrepancies; these are the main candidates for manual curation in subsequent steps.

The following transcripts merit manual inspection:

- Prodigal-only ORFs (no support from Glimmer or GeneMark)
 - 1_12 (10136–10408, +): short CDS; absent in both Glimmer and GeneMark.
 - 1_22 (18981–19151, –): short internal ORF within the larger region covered by GeneMark gene 20; absent in Glimmer and GeneMark.
These should be checked for length, overlaps with neighboring genes, and BLAST/domain support before inclusion in the final gene set.
- Genes with partial or conflicting support
 - 1_2 (404–1072, –): Glimmer predicts a longer CDS (404–1156), whereas GeneMark predicts a shorter CDS (404–889) fully contained within the Glimmer/Prodigal model; start/stop positions and N-terminal evidence (BLAST, domains, RBS) should be verified.
 - 1_11 (8658–9938, –): Glimmer extends the CDS to 10445 and GeneMark ends at 10436; the true stop codon and functional domain coverage need to be resolved.
 - 1_15 (12725–12922, –): supported by GeneMark but missing from Glimmer, and short; should be evaluated for coding potential versus spurious ORF.
 - 1_16 (12923–13102, –): supported by Glimmer but not by GeneMark, also relatively short and adjacent to 1_15; merits scrutiny for overlap and independent homology support.

- 1_23 (19384–19998, +): Prodigal predicts a terminal partial CDS, while GeneMark gene 20 extends much further upstream (17863–19998); the true N-terminus and whether this region contains one or multiple genes need to be clarified with homology and domain analysis.

In contrast, genes such as 1_3, 1_4, 1_5, 1_6, 1_7, 1_8, 1_9, 1_10, 1_13, 1_14, 1_17, 1_18, 1_19, 1_20, and 1_21 show complete agreement between Prodigal, Glimmer, and GeneMark and can be treated as high-confidence CDSs requiring only standard functional annotation.

Sequence annotation with PROKKA.

In this stage, the genomic fragment was annotated with an automated prokaryotic annotation pipeline using default bacterial settings. The pipeline was executed on the 20 kb sequence (`frag19.fasta`) with the command `prokka --outdir prokka_frag19 --locustag KAP19 --kingdom Bacteria --rfam frag19.fasta`, after setting the locale (`LC_ALL=C`) and adding the annotation and rRNA-prediction executables to the environment path. The run produced standard output files, including GFF3 (`PROKKA_12212025.gff`), GenBank, and protein/nucleotide FASTA files, which were then parsed to quantify coding sequences and functional assignments.

From the resulting GFF file, 18 protein-coding sequences (CDS) were identified within the fragment, with no tRNA or rRNA genes detected. Among these 18 CDSs, 10 were assigned specific functional product names, whereas 8 were annotated as hypothetical proteins, indicating that roughly half of the predicted proteins currently lack a clear functional characterization in available reference databases.

```
export LC_ALL=C
/home/Shared/bin/prokka/bin/prokka --help
PATH=$PATH:/home/Shared/bin/prokka/bin
export PATH=$PATH:/home/Shared/tools/barrnap/barrnap/bin/
prokka --outdir prokka_frag19 --locustag KAP19 --kingdom Bacteria --rfam
fragment_19.fasta
cd prokka_frag19

# Number of CDS
grep -c $'\tCDS\t' PROKKA_12212025.gff
# 18

# Number of tRNAs and rRNAs
grep -c $'\ttRNA\t' PROKKA_12212025.gff
# 0
grep -c $'\trRNA\t' PROKKA_12212025.gff
# 0

# CDS whose product is exactly "hypothetical protein"
grep $'\tCDS\t' PROKKA_12212025.gff | grep -c 'product=hypothetical
protein'
```



```
# 8
```

```
grep '$\tCDS\t' PROKKA_12212025.gff | grep 'product=hypothetical  
protein'  
grep '$\tCDS\t' PROKKA_12212025.gff | grep -v 'product=hypothetical  
protein' | wc -l
```

```
# 10
```

Identifying operons.

For each pair of *adjacent* genes on the same strand:

$$\text{intergenic distance} = \text{startdownstream} - \text{endupstream} - 1$$

Operons in genome order (positions increasing):

Operon 1: KAP19_00002–KAP19_00007

Genes and coordinates:

- KAP19_00002: 1297–2736, +, hypothetical protein
- KAP19_00003: 2733–3641, +, LpxP (lipid A palmitoleoyltransferase)
- KAP19_00004: 3634–3951, +, DNA base-flipping protein (AtI)
- KAP19_00005: 3961–4419, +, putative pre-16S rRNA nuclease (YrrK)
- KAP19_00006: 4421–5713, +, hypothetical protein
- KAP19_00007: 5728–6894, +, FdtB
(dTDP-3-amino-3,6-dideoxy- α -D-galactopyranose transaminase)

Intergenic distances (all < 100 bp, several overlaps): –4, –8, 9, 1, 14 bp.

Interpretation: A six-gene operon combining enzymes for lipid A modification (LpxP) and dTDP-sugar biosynthesis (FdtB) with DNA/RNA-related and hypothetical proteins, likely part of a cell-envelope and sugar-modification locus.

Operon 2: KAP19_00014–KAP19_00018

Genes and coordinates:

- KAP19_00014: 13175–13918, +, hypothetical protein
- KAP19_00015: 14039–14491, +, hypothetical protein
- KAP19_00016: 14524–14976, +, 50S ribosomal protein L9 (RpII)
- KAP19_00017: 15038–15943, +, epimerase family protein
- KAP19_00018: 15936–17750, +, 23S rRNA 5-hydroxycytidine synthase (RIhA)

Intergenic distances: 120 bp (00014–00015), 47 bp (00015–00016), 61 bp (00016–00017), –8 bp (00017–00018).

The <100 bp rule KAP19_00015–KAP19_00018 form a four-gene operon. I accepted 120 bp and treated all five as one block.

Interpretation: An operon enriched in translation-related and RNA-modifying functions (RplI, RlhA) plus an epimerase and one hypothetical protein, suggesting a locus involved in ribosome function / RNA modification and associated metabolism.

Operon 3: KAP19_00008–KAP19_00010

Genes and coordinates:

- KAP19_00008: 6961–7473, –, bacterial non-heme ferritin (FtnA)
- KAP19_00009: 7531–8682, –, hypothetical protein
- KAP19_00010: 8658–10436, –, ion-translocating oxidoreductase complex subunit B (RsxB-like)
- KAP19_00011 (10526–11302, –): hypothetical protein;

Interpretation: Four-gene operon encoding an iron-storage protein (ferritin) and a membrane-associated ion-translocating oxidoreductase subunit, plus two hypothetical proteins, likely involved in redox and metal homeostasis.

Operon 4: KAP19_00012–KAP19_00013

Genes and coordinates:

- KAP19_00012: 11427–12728, –, hypothetical protein.
- KAP19_00013: 12725–12922, –, hypothetical protein.

The two CDSs are co-directional on the minus strand and overlap by 4 bp, satisfying the <100 bp intergenic distance criterion for an operon.

Singleton / non-operonic genes

Using the same <100 bp, same-strand rule, the remaining CDSs are singletons:

- KAP19_00001 (404–1156, –): hypothetical protein.

BLAST.

Protein sequences for all 18 Prokka-annotated CDSs were queried using BLASTp against a public protein database to identify homologs and support functional assignments. For each query protein, the top-scoring matches were inspected and recorded (including E-value, percent identity, and query coverage), and the taxonomic distribution of the best hits was used as supporting evidence during functional curation and as an initial screen for potential HGT candidates.

CDSs predicted by Prodigal, but unsupported by any other tool (12, 22) were also queried using BLASTp but didn't produce any significant hits.

Horizontal Gene Transfer Analysis (HGT) using BLASTp, MAFFT, NJ phylogenetic tree.

In this step, potential horizontally transferred genes were investigated by combining BLASTp searches with phylogenetic analysis of selected candidates. Protein sequences of all 18

Prokka-annotated CDSs were queried against public protein databases using BLASTp, and for each gene the taxonomic composition of top hits was inspected to identify cases where high-confidence homologs were predominantly found in lineages distantly related to Candidatus Kapaibacteriota. Most proteins showed best hits to FCB/CFB-group or closely related taxa, and thus were considered unlikely HGT candidates. A [FeFe] hydrogenase large subunit C-terminal domain-containing protein (Prokka ID PROKKA_10) was selected as a putative HGT candidate based on its functional category and broad distribution of homologs. Homologous protein sequences (representatives from multiple phyla) were downloaded from BLAST, aligned with MAFFT using default amino-acid settings, and a maximum-likelihood tree was inferred with NJ (UPGMA tree), with standard setting in MAFFT.

Transcription terminators prediction using TransTermHP.

Intrinsic (Rho-independent) transcription terminators were assessed downstream of the predicted operons using TransTermHP. TransTermHP identifies candidate intrinsic terminators by scanning genomic DNA for hairpin-forming inverted repeats followed by a U/T-rich tract and assigns each prediction a confidence/score reflecting the likelihood of a true terminator.

For the long operon of interest, the downstream intergenic region (approximately 200 bp after the last CDS) was submitted to a TransTermHP web service, but there were no terminator candidates recorded.

antiSMASH secondary metabolite prediction.

Secondary metabolite biosynthetic potential within the 20 kb fragment was evaluated using the antiSMASH web server, which screens genomic DNA for biosynthetic gene clusters (BGCs) by detecting hallmark biosynthetic enzymes and conserved cluster architectures (e.g., NRPS/PKS-, terpene-, and RiPP-associated loci). The fragment FASTA sequence was submitted using default detection settings, but no BGCs were predicted.

Discussion and conclusions.

A central outcome of this work is that gene content in the 20 kb region is structurally robust (multiple predictors broadly agree), whereas biological interpretation is limited by the novelty of the lineage and incomplete reference databases. The cross-tool agreement between Prodigal, GeneMarkS-2, Glimmer3 and PROKKA for most CDS boundaries supports the conclusion that the majority of predicted genes represent genuine coding sequences rather than algorithm-specific artifacts, while a small set of short or boundary-discordant ORFs (e.g., the two Prodigal-only candidates) are best treated as low-confidence until supported by independent evidence such as conserved domains or strong homology.

Despite limited annotation for many proteins, several loci point to coherent biological themes in this genomic segment, particularly cell-envelope modification, iron/redox homeostasis, and RNA/ribosome-associated functions. The long plus-strand operon (KAP19_00002–KAP19_00007) includes an LpxP-like lipid A palmitoleoyltransferase and an FdtB-like sugar aminotransferase, suggesting an island involved in structural tailoring of envelope components (lipid A and nucleotide-activated sugars), which can affect membrane properties and potentially interactions with the environment. FdtB enzymes (dTDP-sugar aminotransferases) participate in the biosynthesis of unusual deoxy/amino sugars that are typically incorporated into surface glycans or related cell-envelope structures; biochemical/structural studies of pathways producing dTDP-linked amino sugars support this general functional interpretation. The presence of multiple hypothetical proteins inside the same operon indicates that additional, lineage-specific components may contribute to envelope biochemistry, and these hypothetical genes become prime targets for domain-based functional prediction and synteny-based inference. [1, 2]

On the minus strand, the operon containing a bacterial non-heme ferritin (FtnA) together with redox-related proteins is consistent with a functional module involved in metal management and oxidative/redox stress. Bacterial ferritins such as FtnA oxidize and mineralize ferrous iron for intracellular storage, contributing to iron homeostasis and reducing iron-driven oxidative damage; experimental and review literature supports these roles for bacterial ferritin-like proteins. In this context, co-localization with an ion-translocating oxidoreductase complex subunit (annotated as RsxB-like) supports a view of this region as participating in redox-linked energy conservation and/or maintenance of the cellular redox balance, although the exact physiological role cannot be concluded from a single fragment. [3, 4]

Operon structure analysis suggests that this region is organized into a small number of multi-gene transcriptional units plus several apparent single-gene units, consistent with compact bacterial genome organization. The inability to confidently identify a canonical σ^{70} promoter (clear –35/–10 pair) upstream of the long operon and the absence of a strong hairpin+poly-T terminator prediction downstream do not contradict operon existence; rather, they highlight the limitations of transferring promoter/terminator expectations from model bacteria to a distant and poorly characterized phylum and to a short, local genomic context.

HGT screening illustrates the difference between “plausible” and “supported” HGT claims. Although [FeFe]-hydrogenases often have complex evolutionary histories and can be found across diverse bacterial clades, the phylogenetic tree built for the PROKKA_10 hydrogenase placed the query sequence within a clade of *Candidatus Kapaibacteriota* homologs rather than being nested within a clearly foreign phylum, which argues against a recent inter-phylum acquisition for this locus. This case therefore serves as a useful control example: a gene that is broadly distributed (and might look “HGT-like” from broad BLAST diversity) can still show a tree topology consistent with vertical inheritance once close relatives are included.

Finally, the secondary-metabolite analysis yielded a negative result: no biosynthetic gene cluster was detected by antiSMASH in the 20 kb fragment and no obvious NRPS/PKS/terpene/RiPP hallmark genes were apparent from the existing annotation. Because many canonical biosynthetic gene clusters are larger than 20 kb and often require broader genomic context for confident detection, absence of evidence in this window should

be interpreted as “no cluster detected in this fragment” rather than “the organism lacks secondary metabolism.” Overall, the fragment appears to encode a combination of envelope-related enzymes, redox/metal-handling components, and conserved housekeeping functions, with multiple hypothetical proteins representing the primary unresolved component and the most promising direction for deeper manual curation.

References.

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2. UniProt Consortium. dTDP-3-amino-3,6-dideoxy- α -D-galactopyranose transaminase (FdtB) (UniProtKB: Q6T1W6) [Internet]. UniProt. Available from: <https://www.uniprot.org/uniprot/Q6T1W6> (accessed 2025-12-21).
3. UniProt Consortium. Ion-translocating oxidoreductase complex subunit B (RnfB/RsxB) (UniProtKB: P77223) [Internet]. UniProt. Available from: <https://www.uniprot.org/uniprot/P77223/entry> (accessed 2025-12-21).
4. Weeratunga S, Gee CE, Lovell S, Zeng Y, Woodin CL, Rivera M. Two distinct ferritin-like molecules in *Pseudomonas aeruginosa*: the product of the *bfrA* gene is a bacterial ferritin (FtnA) and not a bacterioferritin (Bfr). *Biochemistry*. 2011;50(25):5470–5482. PMID: 21574546. PMCID: PMC3130351.

Supplementing materials.

Fig. 4 BLASTp hits for 18 PROKKA predicted CDSs (other CDS2 predicted by different tools and not supported by PROKKA didn't produce any significant hits).

Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Click the  to see the cluster contents									
 3 member(s), 3 organism(s)	bacteria	slipin family protein [Candidatus Kapaibacterium sp.]	491	491	100%	6e-175	96.80%	250	HOMM4907.1
 2 member(s), 2 organism(s)	bacteria	MAG: slipin family protein [Bacteroidetes/Chlorobi group bacterium C...	401	401	98%	4e-139	77.96%	264	ROL57153.1
 3 member(s), 3 organism(s)	bacteria	MAG: slipin family protein [Bacteroidota bacterium]	399	399	98%	3e-138	76.02%	256	TAL66839.1
 1 member(s), 1 organism(s)	bacteria	slipin family protein [Candidatus Kapaibacterium sp.]	398	398	98%	8e-136	75.92%	279	MFP4370339.1
 3 member(s), 2 organism(s)	bacteria	slipin family protein [Candidatus Kapaibacterium sp.]	397	397	97%	2e-137	78.51%	270	MCOS2504117.1
 1 member(s), 1 organism(s)	bacteria	MAG: hypothetical protein A2X64_01815 [Ignavibacteria bacterium ...	395	395	98%	9e-137	76.23%	251	OGU59256.1
 1 member(s), 1 organism(s)	bacteria	slipin family protein [Candidatus Kapaibacterium sp.]	394	394	96%	2e-136	78.75%	249	MCX7737279.1
 1 member(s), 1 organism(s)	CFB group bacteria	slipin family protein [Bacteroidota bacterium]	394	394	96%	3e-136	77.50%	267	MFC2131645.1
 1 member(s), 1 organism(s)	bacteria	slipin family protein [Candidatus Kapaibacterium sp.]	392	392	98%	1e-135	76.33%	268	MDT3738277.1
 1 member(s), 1 organism(s)	bacteria	slipin family protein [Ignavibacteriota bacterium]	392	392	98%	1e-135	75.51%	267	MBX3044022.1
Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Click the  to see the cluster contents									
 2 member(s), 2 organism(s)	bacteria	GWxTD domain-containing protein [Ignavibacteria bacterium]	970	970	100%	0.0	99.16%	479	MCX8054121.1
 2 member(s), 1 organism(s)	bacteria	GWxTD domain-containing protein [Candidatus Kapaibacterium sp.]	853	853	95%	0.0	89.01%	455	HPP38899.1
 1 member(s), 1 organism(s)	bacteria	GWxTD domain-containing protein [Candidatus Kapaibacterium sp.]	711	711	82%	0.0	85.32%	365	HOK15000.1
 1 member(s), 1 organism(s)	bacteria	GWxTD domain-containing protein [Ignavibacteria bacterium]	370	370	38%	2e-124	95.65%	184	MCX8055875.1
 1 member(s), 1 organism(s)	CFB group bacteria	GWxTD domain-containing protein [Bacteroidota bacterium]	351	351	87%	1e-112	41.25%	485	MFA6570800.1
 1 member(s), 1 organism(s)	bacteria	GWxTD domain-containing protein [CFB group bacterium]	347	347	100%	4e-111	38.80%	484	MAG2290088.1
 1 member(s), 1 organism(s)	CFB group bacteria	MAG: GWxTD domain-containing protein [Bacteroidota bacterium]	342	342	95%	6e-109	38.26%	481	TAL68980.1
 1 member(s), 1 organism(s)	bacteria	GWxTD domain-containing protein [Ignavibacteria bacterium]	340	340	96%	2e-108	40.56%	473	NLO20338.1
 1 member(s), 1 organism(s)	bacteria	GWxTD domain-containing protein [Candidatus Kapaibacterium sp.]	322	322	95%	2e-101	36.48%	468	MFP4528177.1
 1 member(s), 1 organism(s)	bacteria	MAG: hypothetical protein A2X63_14240 [Ignavibacteria bacterium ...	322	322	91%	3e-101	39.37%	481	OGU13939.1
Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Click the  to see the cluster contents									
 3 member(s), 2 organism(s)	bacteria	lysophospholipid acyltransferase family protein [Ignavibacteria bacter...	609	609	100%	0.0	98.34%	302	MCX8054120.1
 2 member(s), 1 organism(s)	bacteria	lysophospholipid acyltransferase family protein [Candidatus Kapaiba...	526	526	100%	0.0	87.42%	303	HPU23068.1
 1 member(s), 1 organism(s)	bacteria	lysophospholipid acyltransferase family protein [bacterium]	280	280	96%	4e-90	46.74%	292	MFH1051650.1
 3 member(s), 1 organism(s)	bacteria	lysophospholipid acyltransferase family protein [Candidatus Kapaiba...	278	278	96%	3e-89	48.10%	298	MFA7625604.1
 1 member(s), 1 organism(s)	CFB group bacteria	lysophospholipid acyltransferase family protein [Bacteroidota bacteri...	278	278	96%	3e-89	47.06%	293	MFC2130889.1
 1 member(s), 1 organism(s)	bacteria	lysophospholipid acyltransferase family protein [Candidatus Kapaiba...	274	274	96%	1e-87	45.86%	297	HRP02569.1
 1 member(s), 1 organism(s)	bacteria	lysophospholipid acyltransferase family protein [Candidatus Kapaiba...	274	274	92%	1e-87	46.76%	298	GAB137285.1
 1 member(s), 1 organism(s)	bacteria	lysophospholipid acyltransferase family protein [Ignavibacteriota bact...	273	273	96%	3e-87	46.90%	300	MBX3043647.1
 1 member(s), 1 organism(s)	bacteria	MAG: hypothetical protein D9V86_05230 [Bacteroidetes/Chlorobi gr...	271	271	94%	9e-87	46.23%	297	ROL61543.1
 2 member(s), 2 organism(s)	bacteria	lysophospholipid acyltransferase family protein [Candidatus Kapaiba...	268	268	96%	2e-85	45.70%	296	MBE2189286.1
Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Click the  to see the cluster contents									
 5 member(s), 3 organism(s)	bacteria	MGMT family protein [Ignavibacteria bacterium]	209	209	100%	5e-68	94.29%	105	MCX8055873.1
 1 member(s), 1 organism(s)	bacteria	MGMT family protein [Candidatus Kapaibacteriales bacterium]	138	138	97%	7e-40	62.14%	110	MGQ9819635.1
 2 member(s), 2 organism(s)	CFB group bacteria	MAG: family protein [Flavobacterium sp. UBA4197]	129	129	99%	2e-36	57.14%	110	WP_286912912.1
 1 member(s), 1 organism(s)	bacteria	MGMT family protein [Candidatus Kapaibacterium sp.]	129	129	96%	2e-36	55.45%	105	MDT3738521.1
 1 member(s), 1 organism(s)	bacteria	MGMT family protein [Ignavibacteria bacterium]	129	129	99%	6e-36	57.01%	127	MBD1210129.1
 1 member(s), 1 organism(s)	bacteria	MGMT family protein [Candidatus Kapaibacterium sp.]	128	128	96%	8e-36	58.65%	128	MC96807246.1
 1 member(s), 1 organism(s)	CFB group bacteria	MGMT family protein [Bacteroidota bacterium]	127	127	98%	2e-35	54.37%	120	MCL5992506.1
 1 member(s), 1 organism(s)	bacteria	MGMT family protein [Candidatus Kapaibacteriota bacterium]	126	126	97%	2e-35	55.88%	102	MG9701981.1
 1 member(s), 1 organism(s)	bacteria	MGMT family protein [Candidatus Kapaibacterium sp.]	126	126	96%	4e-35	55.45%	104	MDX9790676.1
 1 member(s), 1 organism(s)	bacteria	MGMT family protein [Candidatus Kapaibacterium sp.]	126	126	97%	4e-35	58.82%	110	MCX7738617.1
Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Click the  to see the cluster contents									
 3 member(s), 2 organism(s)	bacteria	Holliday junction resolvase RuvX [Ignavibacteria bacterium]	299	299	100%	4e-102	96.05%	152	MCX8054118.1
 1 member(s), 1 organism(s)	bacteria	Holliday junction resolvase RuvX [Candidatus Kapaibacterium sp.]	273	273	100%	4e-92	84.87%	152	HOK14409.1
 2 member(s), 2 organism(s)	bacteria	Holliday junction resolvase RuvX [Candidatus Kapaibacterium sp.]	144	144	96%	7e-41	51.37%	148	MCOS252105.1
 1 member(s), 1 organism(s)	bacteria	MAG: Holliday junction resolvase RuvX [Bacteroidetes/Chlorobi grou...	140	140	89%	1e-39	50.74%	147	ROL61034.1
 1 member(s), 1 organism(s)	CFB group bacteria	Holliday junction resolvase RuvX [Bacteroidota bacterium]	137	137	97%	3e-38	48.30%	152	MCL5992125.1
 1 member(s), 1 organism(s)	bacteria	Holliday junction resolvase RuvX [Candidatus Kapaibacterium sp.]	137	137	97%	6e-38	45.58%	153	MFP4529359.1
 1 member(s), 1 organism(s)	bacteria	Holliday junction resolvase RuvX [bacterium]	134	134	93%	4e-37	45.77%	148	MFH1049509.1
 1 member(s), 1 organism(s)	bacteria	Holliday junction resolvase RuvX [Candidatus Kapaibacterium sp.]	134	134	97%	6e-37	48.65%	148	MBE2188235.1
 1 member(s), 1 organism(s)	CFB group bacteria	Holliday junction resolvase RuvX [Bacteroidota bacterium]	134	134	96%	8e-37	47.33%	153	MBQ6269462.1
 1 member(s), 1 organism(s)	bacteria	Holliday junction resolvase RuvX [Candidatus Kapaibacteriales bact...	132	132	90%	4e-36	48.72%	148	MGQ9819692.1
Cluster Composition	Cluster Ancestor	Cluster Representative Sequence	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Click the  to see the cluster contents									
 2 member(s), 2 organism(s)	bacteria	GWxTD domain-containing protein [Ignavibacteria bacterium]	856	856	100%	0.0	97.21%	430	MCX8054117.1
 1 member(s), 1 organism(s)	bacteria	GWxTD domain-containing protein [Candidatus Kapaibacterium sp.]	702	702	100%	0.0	77.44%	430	HOM04902.1
 1 member(s), 1 organism(s)	bacteria	GWxTD domain-containing protein [Candidatus Kapaibacterium sp.]	607	607	85%	0.0	79.12%	364	HOK049440.1
 1 member(s), 1 organism(s)	bacteria	hypothetical protein [Ignavibacteria bacterium]	605	605	81%	0.0	84.77%	348	MCX8055871.1
 1 member(s), 1 organism(s)	bacteria	hypothetical protein [Candidatus Kapaibacterium sp.]	491	491	74%	9e-171	73.44%	320	HOK14408.1
 1 member(s), 1 organism(s)	bacteria	GWxTD domain-containing protein [Candidatus Kapaibacterium sp.]	218	218	99%	4e-62	34.15%	455	MDT3740378.1
 1 member(s), 1 organism(s)	CFB group bacteria	GWxTD domain-containing protein [Bacteroidota bacterium]	217	217	92%	8e-62	33.25%	455	MCL5992126.1
 2 member(s), 1 organism(s)	bacteria	GWxTD domain-containing protein [Candidatus Kapaibacterium sp.]	208	208	90%	2e-58	33.57%	452	MFA5512478.1
 1 member(s), 1 organism(s)	bacteria	hypothetical protein MASR1M45_11900 [Candidatus Kapaibacterium s...	202	202	98%	6e-56	32.52%	450	GAB1371128.1
 1 member(s), 1 organism(s)	bacteria	GWxTD domain-containing protein [Candidatus Kapaibacterium sp.]	201	201	91%	8e-56	32.10%	452	MFP4528358.1

	Cluster Composition ▲ Click the ID to see the cluster contents	Cluster Ancestor ▼	Cluster Representative Sequence ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
✓	1 member(s), 3 organism(s)	bacteria	DegT/DnrJ/EryC1/StrS family aminotransferase [Candidatus Kapaibacte...	706	706	95%	0.0	90.79%	369	HOK14195.1
✓	1 member(s), 1 organism(s)	bacteria	DegT/DnrJ/EryC1/StrS family aminotransferase [Candidatus Kapaibacte...	583	583	100%	0.0	70.10%	388	MFP4527195.1
✓	1 member(s), 1 organism(s)	bacteria	MAG: transcripional regulator [Ignavibacteria bacterium GWA2_35_8]	583	583	100%	0.0	70.80%	387	OGU17886.1
✓	1 member(s), 1 organism(s)	CFB group bacteria	DegT/DnrJ/EryC1/StrS family aminotransferase [Bacteroidota bacterium]	582	582	99%	0.0	69.87%	387	MFC2130810.1
✓	2 member(s), 2 organism(s)	bacteria	MAG: DegT/DnrJ/EryC1/StrS family aminotransferase [Bacteroidetes/C...	578	578	99%	0.0	69.61%	387	ROL60651.1
✓	2 member(s), 2 organism(s)	bacteria	DegT/DnrJ/EryC1/StrS family aminotransferase [Bacteroidota bacterium]	577	577	100%	0.0	69.77%	387	MES2784730.1
✓	1 member(s), 1 organism(s)	GNS bacteria	DegT/DnrJ/EryC1/StrS family aminotransferase [Chloroflexota bacterium]	577	577	99%	0.0	72.06%	384	MGE5479235.1
✓	1 member(s), 1 organism(s)	CFB group bacteria	MAG: DegT/DnrJ/EryC1/StrS family aminotransferase [Bacteroidota bac...	574	574	100%	0.0	68.81%	388	TAL67690.1
✓	2 member(s), 2 organism(s)	bacteria	DegT/DnrJ/EryC1/StrS family aminotransferase [Candidatus Kapaibacte...	573	573	100%	0.0	71.58%	388	MFN8362029.1
✓	1 member(s), 1 organism(s)	bacteria	DegT/DnrJ/EryC1/StrS family aminotransferase [Ignavibacteriota bacteri...	572	572	99%	0.0	69.17%	387	MBJ5324696.1
	Cluster Composition ▲ Click the ID to see the cluster contents	Cluster Ancestor ▼	Cluster Representative Sequence ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
✓	2 member(s), 2 organism(s)	bacteria	ferritin [Candidatus Kapaibacterium sp.]	344	344	100%	2e-119	98.24%	170	HOK14706.1
✓	1 member(s), 1 organism(s)	bacteria	ferritin [Candidatus Kapaibacterium sp.]	301	301	88%	1e-102	98.00%	150	HPU24019.1
✓	1 member(s), 1 organism(s)	bacteria	ferritin [Candidatus Kapaibacterium sp.]	289	289	86%	7e-98	96.58%	146	HOQ48654.1
✓	2 member(s), 1 organism(s)	bacteria	ferritin [Candidatus Kapaibacterium sp.]	274	274	95%	8e-92	79.50%	168	MFP4369416.1
✓	1 member(s), 1 organism(s)	bacteria	Ferroxidase [Ignavibacteria bacterium]	246	246	99%	1e-80	67.86%	170	MBM2813839.1
✓	1 member(s), 1 organism(s)	bacteria	ferritin [Candidatus Kapaibacterium sp.]	244	244	99%	1e-79	66.86%	172	MCX6154247.1
✓	1 member(s), 1 organism(s)	CFB group bacteria	ferritin [Bacteroidota bacterium]	238	238	97%	1e-77	67.27%	171	MAF6569906.1
✓	1 member(s), 1 organism(s)	GNS bacteria	ferritin [Chloroflexota bacterium]	236	236	97%	1e-76	65.45%	170	MGE5479623.1
✓	1 member(s), 1 organism(s)	bacteria	ferritin [Candidatus Kapaibacteriota bacterium]	235	235	95%	2e-76	68.32%	173	MG89777055.1
✓	3 member(s), 3 organism(s)	bacteria	ferritin [Candidatus Kapaibacteriota bacterium]	233	233	97%	1e-75	66.06%	165	MFN3269022.1
	Cluster Composition ▲ Click the ID to see the cluster contents	Cluster Ancestor ▼	Cluster Representative Sequence ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
✓	1 member(s), 3 organism(s)	bacteria	serine/threonine-protein phosphatase [Ignavibacteria bacterium]	781	781	100%	0.0	99.74%	383	MCX8054114.1
✓	1 member(s), 1 organism(s)	bacteria	SpoIIIE family protein phosphatase [Candidatus Kapaibacterium sp.]	669	669	86%	0.0	98.18%	330	HOK14705.1
✓	1 member(s), 1 organism(s)	bacteria	PP2C family protein-serine/threonine phosphatase [Candidatus Kap...	483	483	63%	1e-169	97.93%	241	HPP39408.1
✓	1 member(s), 1 organism(s)	bacteria	SpoIIIE family protein phosphatase [Candidatus Kapaibacterium sp.]	407	407	99%	2e-137	51.97%	382	MFP43689335.1
✓	1 member(s), 1 organism(s)	bacteria	SpoIIIE family protein phosphatase [bacterium]	397	397	99%	1e-133	49.21%	382	MFH0733361.1
✓	1 member(s), 1 organism(s)	bacteria	SpoIIIE family protein phosphatase [Ignavibacteriaceae bacterium]	397	397	99%	2e-133	51.45%	382	HPJ38647.1
✓	1 member(s), 1 organism(s)	bacteria	SpoIIIE family protein phosphatase [Ignavibacteria bacterium]	392	392	99%	2e-131	49.08%	384	HPS64969.1
✓	1 member(s), 1 organism(s)	bacteria	SpoIIIE family protein phosphatase [Armatimonadota bacterium]	391	391	99%	4e-131	46.72%	384	NLB66627.1
✓	3 member(s), 3 organism(s)	bacteria	SpoIIIE family protein phosphatase [Ignavibacteria bacterium]	386	386	99%	4e-129	49.47%	382	MBN8545273.1
✓	3 member(s), 3 organism(s)	bacteria	SpoIIIE family protein phosphatase [Armatimonadota bacterium]	385	385	99%	5e-129	48.09%	384	MBN1036904.1
	Cluster Composition ▲ Click the ID to see the cluster contents	Cluster Ancestor ▼	Cluster Representative Sequence ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
✓	3 member(s), 3 organism(s)	bacteria	[Fe-Fe] hydrogenase large subunit C-terminal domain-containing pro...	1124	1124	100%	0.0	91.72%	591	HOM04101.1
✓	1 member(s), 1 organism(s)	bacteria	[Fe-Fe] hydrogenase large subunit C-terminal domain-containing pro...	895	895	81%	0.0	91.02%	478	HOK15135.1
✓	2 member(s), 1 organism(s)	bacteria	[Fe-Fe] hydrogenase large subunit C-terminal domain-containing pro...	651	651	99%	0.0	51.19%	591	MEN6844973.1
✓	1 member(s), 1 organism(s)	bacteria	[Fe-Fe] hydrogenase large subunit C-terminal domain-containing pro...	639	639	100%	0.0	53.03%	584	MFP4368938.1
✓	3 member(s), 1 organism(s)	bacteria	[Fe-Fe] hydrogenase large subunit C-terminal domain-containing pro...	638	638	100%	0.0	52.37%	581	MEQ8167686.1
✓	1 member(s), 1 organism(s)	bacteria	[Fe-Fe] hydrogenase large subunit C-terminal domain-containing pro...	636	636	99%	0.0	50.43%	591	MFA7418373.1
✓	1 member(s), 1 organism(s)	bacteria	4Fe-4S binding protein [Physphaerae bacterium]	636	636	99%	0.0	51.18%	598	NLE30247.1
✓	1 member(s), 1 organism(s)	bacteria	[Fe-Fe] hydrogenase large subunit C-terminal domain-containing pro...	633	633	99%	0.0	50.76%	590	MDD5361833.1
✓	1 member(s), 1 organism(s)	bacteria	4Fe-4S binding protein [bacterium]	633	633	99%	0.0	50.59%	592	MCE5240931.1
✓	1 member(s), 1 organism(s)	bacteria	[Fe-Fe] hydrogenase large subunit C-terminal domain-containing pro...	632	632	99%	0.0	50.08%	592	MEI6501384.1
	Cluster Composition ▲ Click the ID to see the cluster contents	Cluster Ancestor ▼	Cluster Representative Sequence ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
✓	2 member(s), 2 organism(s)	bacteria	DUF481 domain-containing protein [Ignavibacteria bacterium]	523	523	100%	0.0	100.00%	258	MCX8054112.1
✓	2 member(s), 1 organism(s)	bacteria	DUF481 domain-containing protein [Candidatus Kapaibacterium sp.]	448	448	98%	1e-157	88.98%	261	HOK15314.1
✓	1 member(s), 1 organism(s)	bacteria	DUF481 domain-containing protein [Candidatus Zhupueibacteriota ba...	183	183	93%	4e-53	40.25%	262	MBN1351635.1
✓	1 member(s), 1 organism(s)	bacteria	DUF481 domain-containing protein [Candidatus Glassiibacteriota bac...	179	179	98%	1e-51	35.18%	253	MFO6104494.1
✓	1 member(s), 1 organism(s)	green sulfur bacteria	DUF481 domain-containing protein [Chlorobiota bacterium]	178	178	98%	2e-51	37.94%	255	MFN3196174.1
✓	2 member(s), 1 organism(s)	bacteria	DUF481 domain-containing protein [Candidatus Kapaibacterium sp.]	168	168	99%	2e-47	37.11%	253	MFP4370408.1
✓	1 member(s), 1 organism(s)	bacteria	DUF481 domain-containing protein [Candidatus Zhupueibacteriota ba...	166	166	95%	1e-46	37.55%	261	MBD32809663.1
✓	1 member(s), 1 organism(s)	green sulfur bacteria	DUF481 domain-containing protein [Chlorobiota bacterium]	163	163	100%	4e-45	33.20%	269	MBL7886458.1
✓	1 member(s), 1 organism(s)	bacteria	DUF481 domain-containing protein [Armatimonadota bacterium]	162	162	98%	5e-45	32.68%	265	MCC7438816.1
✓	1 member(s), 1 organism(s)	bacteria	DUF481 domain-containing protein [Fideliibacteriota bacterium]	161	161	98%	1e-44	33.86%	263	MBD3276348.1
	Cluster Composition ▲ Click the ID to see the cluster contents	Cluster Ancestor ▼	Cluster Representative Sequence ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
✓	1 member(s), 1 organism(s)	CFB group bacteria	DUF6588 family protein [Bacteroidota bacterium]	865	865	100%	0.0	98.61%	433	MEJ5244912.1
✓	2 member(s), 1 organism(s)	bacteria	hypothetical protein [Candidatus Kapaibacterium sp.]	793	793	100%	0.0	87.76%	433	HOM04103.1
✓	1 member(s), 1 organism(s)	bacteria	hypothetical protein [Ignavibacteria bacterium]	569	569	65%	0.0	97.53%	283	MCX8064111.1
✓	1 member(s), 1 organism(s)	bacteria	hypothetical protein [Ignavibacteria bacterium]	431	431	100%	2e-145	49.89%	428	NLO19886.1
✓	1 member(s), 1 organism(s)	bacteria	DUF6588 family protein [Candidatus Kapaibacterium sp.]	419	419	100%	7e-141	49.89%	415	MFP4527353.1
✓	1 member(s), 1 organism(s)	bacteria	hypothetical protein [Candidatus Kapaibacterium sp.]	410	410	100%	4e-137	46.22%	448	MDD3125301.1
✓	1 member(s), 1 organism(s)	bacteria	hypothetical protein [Candidatus Kapaibacterium sp.]	410	410	100%	4e-137	46.24%	440	MDA3844804.1
✓	1 member(s), 1 organism(s)	bacteria	hypothetical protein [Ignavibacteriota bacterium]	407	407	100%	4e-136	48.52%	425	MBX3042272.1
✓	1 member(s), 1 organism(s)	CFB group bacteria	DUF6588 family protein [Bacteroidota bacterium]	407	407	93%	4e-136	48.40%	426	MFA6569715.1
✓	1 member(s), 1 organism(s)	bacteria	DUF6588 family protein [Candidatus Kapaibacterium sp.]	407	407	100%	8e-136	48.06%	427	MDT3739073.1

	Cluster Composition ▲ Click the ▲ to see the cluster contents	Cluster Ancestor ▼	Cluster Representative Sequence ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
✓	1 member(s), 1 organism(s)	CFB group bacteria	hypothetical protein [Bacteroidota bacterium]	126	126	100%	2e-36	100.00%	65	MEJ5244913.1
	Cluster Composition ▲ Click the ▲ to see the cluster contents	Cluster Ancestor ▼	Cluster Representative Sequence ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
✓	1 member(s), 1 organism(s)	CFB group bacteria	DUF4292 domain-containing protein [Bacteroidota bacterium]	476	476	100%	3e-169	95.95%	247	MEJ5244915.1
✓	2 member(s), 1 organism(s)	bacteria	DUF4292 domain-containing protein [Candidatus Kapaibacterium sp.]	379	379	100%	2e-130	76.11%	249	HOQ49012.1
✓	1 member(s), 1 organism(s)	bacteria	DUF4292 domain-containing protein [Candidatus Kapaibacterium sp.]	137	137	98%	1e-35	29.80%	252	MFP4528742.1
✓	1 member(s), 1 organism(s)	bacteria	DUF4292 domain-containing protein [Candidatus Kapaibacterium sp.]	133	133	99%	8e-34	31.73%	255	MDT3740930.1
✓	1 member(s), 1 organism(s)	bacteria	DUF4292 domain-containing protein [Ignavibacteriota bacterium]	126	126	94%	2e-31	34.04%	256	MBX3044361.1
✓	2 member(s), 1 organism(s)	bacteria	DUF4292 domain-containing protein [Candidatus Kapaibacterium sp.]	124	124	94%	2e-30	31.62%	257	MDX9790864.1
✓	1 member(s), 1 organism(s)	CFB group bacteria	DUF4292 domain-containing protein [Bacteroidota bacterium]	117	117	100%	7e-28	29.60%	250	MFA6570989.1
✓	1 member(s), 1 organism(s)	bacteria	DUF4292 domain-containing protein [Candidatus Kapaibacterium sp.]	115	115	97%	6e-27	28.10%	247	MDQ1067732.1
✓	1 member(s), 1 organism(s)	bacteria	hypothetical protein [Ignavibacteria bacterium]	112	112	96%	2e-25	29.75%	324	MBM2815041.1
✓	1 member(s), 1 organism(s)	bacteria	MAG: hypothetical protein CV22_02040 [Ignavibacteriae bacterium...]	110	110	100%	3e-25	30.65%	250	PKL88893.1
	Cluster Composition ▲ Click the ▲ to see the cluster contents	Cluster Ancestor ▼	Cluster Representative Sequence ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
✓	3 member(s), 2 organism(s)	bacteria	30S ribosomal protein S6 [Candidatus Kapaibacterium sp.]	291	291	100%	5e-99	95.33%	150	HOK14006.1
✓	1 member(s), 1 organism(s)	GNS bacteria	30S ribosomal protein S6 [Chloroflexota bacterium]	205	205	94%	5e-65	65.03%	159	MGE5480680.1
✓	2 member(s), 2 organism(s)	bacteria	30S ribosomal protein S6 [Ignavibacteria bacterium]	197	197	81%	1e-61	76.23%	181	NLJ15466.1
✓	2 member(s), 1 organism(s)	bacteria	30S ribosomal protein S6 [Candidatus Kapaibacterium sp.]	194	194	93%	9e-61	66.43%	152	MDX9789455.1
✓	1 member(s), 1 organism(s)	bacteria	30S ribosomal protein S6 [Candidatus Kapaibacterium sp.]	194	194	93%	1e-60	65.71%	143	HRP02001.1
✓	1 member(s), 1 organism(s)	bacteria	30S ribosomal protein S6 [Candidatus Kapaibacterium sp.]	193	193	93%	3e-60	64.03%	159	MGD3125962.1
✓	1 member(s), 1 organism(s)	bacteria	30S ribosomal protein S6 [Candidatus Kapaibacterium sp.]	191	191	95%	2e-59	60.81%	155	MFP4368870.1
✓	1 member(s), 1 organism(s)	bacteria	30S ribosomal protein S6 [Candidatus Kapaibacterium sp.]	189	189	87%	6e-59	68.46%	143	MDT3738915.1
✓	1 member(s), 1 organism(s)	CFB group bacteria	30S ribosomal protein S6 [Bacteroidota bacterium]	189	189	93%	1e-58	63.12%	151	MFA6572150.1
✓	1 member(s), 1 organism(s)	bacteria	30S ribosomal protein S6 [Candidatus Kapaibacterium sp.]	189	189	84%	1e-58	68.18%	146	GAB1372114.1
	Cluster Composition ▲ Click the ▲ to see the cluster contents	Cluster Ancestor ▼	Cluster Representative Sequence ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
✓	2 member(s), 2 organism(s)	bacteria	50S ribosomal protein L9 [Candidatus Kapaibacterium sp.]	296	296	100%	4e-101	98.67%	150	HOK14005.1
✓	1 member(s), 1 organism(s)	bacteria	50S ribosomal protein L9 [Candidatus Kapaibacterium sp.]	204	204	100%	2e-64	63.33%	150	MGJ6370601.1
✓	1 member(s), 1 organism(s)	bacteria	50S ribosomal protein L9 [Ignavibacteria bacterium]	199	199	99%	1e-62	62.42%	149	NLO18481.1
✓	1 member(s), 1 organism(s)	bacteria	50S ribosomal protein L9 [Candidatus Kapaibacterium sp.]	198	198	99%	3e-62	59.46%	148	MBL7998903.1
✓	1 member(s), 1 organism(s)	bacteria	50S ribosomal protein L9 [Candidatus Kapaibacterium sp.]	196	196	99%	1e-61	58.78%	148	MBL7975941.1
✓	1 member(s), 1 organism(s)	CFB group bacteria	large subunit ribosomal protein [Bacteroidota bacterium]	194	194	99%	7e-61	58.78%	148	MDQ1267141.1
✓	1 member(s), 1 organism(s)	bacteria	50S ribosomal protein L9 [Candidatus Kapaibacterium sp.]	194	194	99%	8e-61	59.46%	148	MCU0331463.1
✓	3 member(s), 3 organism(s)	bacteria	50S ribosomal protein L9 [Bacteroidota bacterium]	194	194	100%	8e-61	57.33%	150	HAU08275.1
✓	3 member(s), 3 organism(s)	bacteria	50S ribosomal protein L9 [Desulfobulbaceae bacterium]	194	194	99%	1e-60	60.81%	149	MBS3999257.1
✓	2 member(s), 1 organism(s)	bacteria	50S ribosomal protein L9 [Candidatus Kapaibacterium sp.]	194	194	99%	1e-60	59.06%	149	HPJ20652.1
	Cluster Composition ▲ Click the ▲ to see the cluster contents	Cluster Ancestor ▼	Cluster Representative Sequence ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
✓	2 member(s), 2 organism(s)	bacteria	TIGR01777 family oxidoreductase [Bacteroidota bacterium]	603	603	100%	0.0	97.34%	301	MEJ5244918.1
✓	1 member(s), 1 organism(s)	bacteria	TIGR01777 family oxidoreductase [Candidatus Kapaibacterium sp.]	526	526	100%	0.0	81.73%	301	HOK14004.1
✓	1 member(s), 1 organism(s)	bacteria	TIGR01777 family oxidoreductase [Candidatus Kapaibacterium sp.]	475	475	88%	1e-167	83.08%	266	HPU24291.1
✓	1 member(s), 1 organism(s)	bacteria	TIGR01777 family oxidoreductase [Candidatus Kapaibacterium sp.]	414	414	81%	1e-143	79.42%	243	HPP40788.1
✓	2 member(s), 1 organism(s)	bacteria	TIGR01777 family oxidoreductase [Rhodohalobacter sp.]	278	278	98%	4e-89	46.64%	300	WYP_322572735.1
✓	1 member(s), 1 organism(s)	bacteria	TIGR01777 family oxidoreductase [Candidatus Zhuqueibacterota ba...]	275	275	99%	4e-88	44.55%	303	MCA9731150.1
✓	1 member(s), 1 organism(s)	bacteria	TIGR01777 family oxidoreductase [Balneolales bacterium]	273	273	99%	4e-87	45.07%	298	MCH8568455.1
✓	1 member(s), 1 organism(s)	bacteria	MAG: TIGR01777 family protein [bacterium]	271	271	98%	1e-86	43.58%	295	TNE74406.1
✓	1 member(s), 1 organism(s)	CFB group bacteria	TIGR01777 family oxidoreductase [Bacteroidota bacterium]	271	271	99%	3e-86	42.57%	303	MDP1677310.1
✓	2 member(s), 2 organism(s)	bacteria	TIGR01777 family protein [Balneola sp.]	270	270	99%	4e-86	44.22%	299	MAO63722.1
	Cluster Composition ▲ Click the ▲ to see the cluster contents	Cluster Ancestor ▼	Cluster Representative Sequence ▼	Max Score ▼	Total Score ▼	Query Cover ▼	E value ▼	Per. Ident ▼	Acc. Len ▼	Accession
✓	2 member(s), 2 organism(s)	bacteria	U32 family peptidase [Ignavibacteria bacterium]	1210	1210	100%	0.0	98.51%	604	MCX8055450.1
✓	3 member(s), 1 organism(s)	bacteria	U32 family peptidase [Candidatus Kapaibacterium sp.]	1017	1017	100%	0.0	81.46%	603	HOM04109.1
✓	1 member(s), 1 organism(s)	bacteria	U32 family peptidase [Candidatus Kapaibacterium sp.]	822	822	76%	0.0	85.25%	461	HOQ49637.1
✓	1 member(s), 1 organism(s)	bacteria	U32 family peptidase [Ignavibacteriales bacterium]	570	570	99%	0.0	48.76%	615	MBI1937179.1
✓	1 member(s), 1 organism(s)	bacteria	U32 family peptidase [Meliobacter sp.]	570	570	99%	0.0	49.27%	611	MBS3944623.1
✓	1 member(s), 1 organism(s)	CFB group bacteria	MAG: hypothetical protein A2W91_00540 [Bacteroidetes bacterium GW...]	569	569	98%	0.0	50.00%	611	OFY36907.1
✓	1 member(s), 1 organism(s)	bacteria	MAG: U32 family peptidase [Ignavibacteriales bacterium]	569	569	99%	0.0	49.17%	605	RJP71258.1
✓	1 member(s), 1 organism(s)	bacteria	U32 family peptidase [Meliobacteriaceae bacterium]	566	566	98%	0.0	47.60%	611	HOJ29196.1
✓	1 member(s), 1 organism(s)	CFB group bacteria	hypothetical protein [Bacteroidales bacterium]	565	565	98%	0.0	48.92%	606	HAF27559.1
✓	1 member(s), 1 organism(s)	CFB group bacteria	U32 family peptidase [Bacteroidales bacterium]	563	563	98%	0.0	49.50%	609	MBJ9054156.1